Fundamentals of Digital Signal Detection in Noise

Digital signal detection in the presence of noise is realized by comparing received signals against thresholds, using methods such as correlation, matched filtering, and likelihood ratio testing. The aim is to maximize the signal-to-noise ratio (SNR) using statistical techniques to distinguish desired information from random noise.

Reading Material: Digital Signal Detection Techniques

Reading Material 1: (Signal Detection Techniques)

Tuzlukov, V. (2018). *Signal Processing Noise.* Ukraine: CRC Press.

`https://www.google.co.ke/books/edition/Signal_Processing_Noise/xffLBQAAQBAJ?hl=en&gbpv=0`

Advanced Digital Signal Techniques

Advanced digital signal detection techniques aim at handling real-world challenges matched such as interference, fading channels, limited bandwidth, and multi-user environments. These techniques are widely used in modern wireless, radar, and communication systems. Watch the video on the advanced signal detection in 4.2.1 make a summary and key highlights then hold an online discussion on different advanced detection techniques and their appllications